

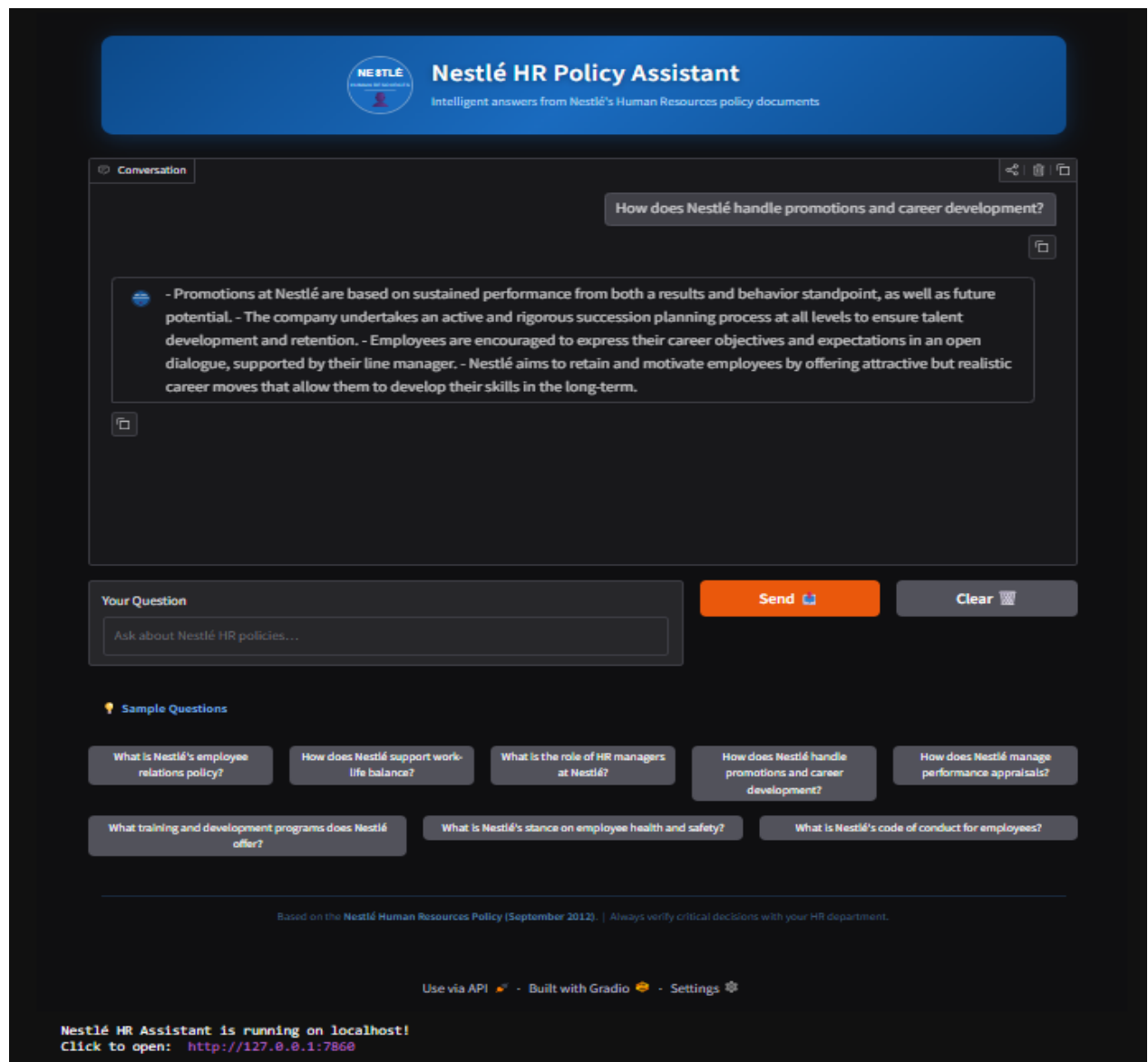
Crafting an AI-Powered HR Assistant

A Use Case for Nestlé's HR Policy Documents Project

Screen Short

1) Gradio Chatbot Interface — Running on Localhost

The final Gradio UI is launched at <http://127.0.0.1:7860> using a dark professional theme with the Nestlé branding. The interface features a conversation panel, a question input box, Send/Clear buttons, and 8 pre-built sample question buttons for quick access. The footer confirms the app is live: "Nestlé HR Assistant is running on localhost!"



2] Load Nestlé HR Policy PDF

The PDF is loaded using PyPDFLoader from LangChain Community. The path is set to THE_NE_1.PDF and the document is split into individual pages. The output confirms 8 pages were loaded successfully, and a sample from Page 1 shows the policy metadata: Policy · Mandatory · September 2012 · The Nestlé Human Resources Policy

Step 4: Load Nestlé HR Policy PDF

Upload the Nestlé HR Policy PDF file and provide the correct path below.

```
[7]: # IMPORTANT: Update this path to your PDF file location
```

```
PDF_PATH = "THE_NE_1.PDF"
```

```
# Load the PDF document
```

```
loader = pdf_loader(PDF_PATH)
```

```
documents = loader.load()
```

```
print(f"PDF loaded successfully!")
```

```
print(f"Total pages loaded: {len(documents)}")
```

```
print(f"\nSample content from Page 1:")
```

```
print(documents[0].page_content[:500])
```

```
PDF loaded successfully!
```

```
Total pages loaded: 8
```

```
Sample content from Page 1:
```

```
Policy
```

```
Mandatory
```

```
September 2012
```

```
The Nestlé
```

```
Human Resources Policy
```

3] Build the Question-Answering System with GPT-3.5 Turbo

The ChatOpenAI model is initialized with model_name="gpt-3.5-turbo", a temperature of 0.2 (for focused, factual answers), and max_tokens=800. This configuration ensures the model stays accurate and avoids hallucination by using a low temperature setting — making it ideal for HR policy Q&A; tasks.

Step 8: Build the Question-Answering System with GPT-3.5 Turbo

```
[7]: # Initialize the GPT-3.5 Turbo language model
```

```
llm = chat_model(
```

```
    model_name="gpt-3.5-turbo",
```

```
    temperature=0.2,          # Lower temperature = more focused, factual answers
```

```
    max_tokens=800
```

```
)
```

```
print("GPT-3.5 Turbo model initialized successfully!")
```

```
GPT-3.5 Turbo model initialized successfully!
```

4] Test the QA System

Three sample HR questions are tested to validate the retrieval and generation pipeline. The system correctly handles policy questions about hiring, training & development, and Total Rewards components. When context is insufficient, it politely redirects the user to consult the HR department - demonstrating responsible AI behaviour.

Step 11: Test the QA System

```
]]: def ask_question(question, session_id="test-session"):
    """Function to query the HR assistant."""
    result = qa_chain.invoke(
        {"question": question},
        config={"configurable": {"session_id": session_id}}
    )

    return result

# Test with sample questions
sample_questions = [
    "What is Nestlé's policy on hiring employees?",
    "How does Nestlé handle employee training and development?",
    "What are the Total Rewards components at Nestlé?"
]

print("-" * 60)
print("TESTING THE NESTLÉ HR ASSISTANT")
print("-" * 60)

for q in sample_questions:
    print(f"Q: {q}")
    print(f"A: {ask_question(q)}")
    print("-" * 60)
```

```
=====
TESTING THE NESTLÉ HR ASSISTANT
=====

Q: What is Nestlé's policy on hiring employees?
A: I'm sorry, I don't have enough information in the Nestlé HR Policy document to answer that question. Please consult the HR department.
-----

Q: How does Nestlé handle employee training and development?
A: - Learning is a part of Nestlé's company culture.
- Employees are encouraged to upgrade their knowledge and skills.
- The company determines training and development priorities.
- Responsibility for implementing these priorities is shared between employees, line managers, and HR.
- Experience and on-the-job training are the primary sources of learning.
- Managers guide and coach employees to succeed in their current positions.
- Continuous improvement, sharing knowledge, and ideas are valued.
- Practices like lateral professional development, extension of responsibilities, and cross-functional teams are encouraged.
- Nestlé offers a comprehensive range of training activities and methodologies to support employee development.
-----

Q: What are the Total Rewards components at Nestlé?
A: - Fixed Pay
- Variable Pay
- Benefits
- Personal Growth and Development
- Work Life Environment
-----
```